

Definition. A subgroup H of G is called a *normal subgroup* of G if: $aH = Ha$ for all $a \in G$.

- left cosets equal right cosets.

Notation: $H \triangleleft G$.

- does not say $ah = ha$! but $ah = h'a$.

- All subgroups of Abelian groups are normal

- $Z(G) \triangleleft G$, for all G .

Theorem (Normal Subgroup Test). $H \triangleleft G$ iff $xHx^{-1} \subseteq H$ $\forall x \in G$.

Proof:

$\Rightarrow$ Let $H \triangleleft G$. Then for all $x \in G$ and $h \in H$, $xh = h'x$ for some $h' \in H$. Thus $xhx^{-1} = h' \in H$, and $xHx^{-1} \subseteq H$.

$\Leftarrow$ Let $xHx^{-1} \subseteq H$, for all $x \in G$.

① Let $a = x$. Then $aHa^{-1} \subseteq H \Rightarrow aH \subseteq Ha$.

② Let $a = x^{-1}$. Then $a^{-1}Ha \subseteq H \Rightarrow Ha \subseteq aH$.

Since $aH \subseteq Ha$ and $Ha \subseteq aH$, $aH = Ha$. Thus $H \triangleleft G$.

Question 1. Let $H = \{(1), (12)(34)\}$ in A_4 . Is H normal in G ? $|A_4| = 12$.

No! $(123)(12)(34) = (134)(2)$ $(123)H = \{(123), (134)\}$
 $(12)(34)(123) = (1)(243)$ $H(123) = \{(123), (243)\}$
 $(123)H \neq H(123)$

Question 2. Is A_n a normal subgroup of S_n ? Why or why not?

yes! There are only 2 cosets of A_n in S_n : A_n and $(1234)A_n$.

Thus $(1234)A_n$ must equal $A_n(1234)$ since $(1234) \notin A_n$.

* True for any $H \subseteq G$ where $[G:H] = 2$.

Question 3. Is $K = \{R_0, H\}$ normal in D_4 ?

No! $R_{90}K = \{R_{90}, D'\} \neq \{R_{90}, D\} = K \cdot R_{90}$

Note: Normality is not Transitive!

Let $J = \{R_0, R_{180}, H, V\}$ Then $K \triangleleft J$ and $J \triangleleft D_4$ but $K \not\triangleleft D_4$.

Question 4. Suppose H is a subgroup of G and H is the only subgroup of G with order k . Why is H normal in G ?

Recall xHx^{-1} is a subgroup of G . Also $|xHx^{-1}| = |H|$.

Thus $xHx^{-1} \subseteq H$, and by normal subgroup test, $H \triangleleft G$.

Exercise 1. Prove that the intersection of two normal subgroups is normal.

Let $H \triangleleft G$ and $K \triangleleft G$. [show $xH \cap Kx^{-1} \subseteq H \cap K$ by showing $xyx^{-1} \in H \cap K$].

Let $y \in H \cap K$. Then $xyx^{-1} \in H$ and $xyx^{-1} \in K$ (b/c H & K are normal).

Thus $xyx^{-1} \in H \cap K$. Therefore $xH \cap Kx^{-1} \subseteq H \cap K$ and by the Normal Subgroup Test, $H \cap K$ is normal in G .

Theorem (Factor Groups). Let G be a group and let H be normal in G . Then the set

$G/H = \{aH : a \in G\}$ is a group under the operation

$$(aH)(bH) = (ab)H.$$

- identity is H

- $(gH)^{-1} = g^{-1}H$.

Question 5. Let $G = \mathbb{Z}_{12}$. Let $H = \langle 3 \rangle$. What are the elements of G/H ?

$H \triangleleft \mathbb{Z}_{12}$ since $\mathbb{Z}_{12}$ is Abelian. $|\langle 3 \rangle| = 4$. Thus $|G/H| = 12/4 = 3$

$$G/H = \{0 + \langle 3 \rangle, 1 + \langle 3 \rangle, 2 + \langle 3 \rangle\} \quad |0 + \langle 3 \rangle| = 1 \quad |1 + \langle 3 \rangle| = 3 \\ |2 + \langle 3 \rangle| = 3$$

Order of Elements in FGs:

$|gH|$ in G/H is the smallest integer k for which $g^k \in H$.

Question 6. Consider the group $G = \mathbb{Z}_{36} / \langle 4 \rangle$. What are the orders of the elements in G ?

$$|0 + \langle 4 \rangle| = 1 \quad |2 + \langle 4 \rangle| = 2$$

$$|1 + \langle 4 \rangle| = 4 \quad |3 + \langle 4 \rangle| = 4$$

22 **Question 7.** Let $G = \mathbb{Z}_8 \oplus \mathbb{Z}_4$ and let $H = \langle (2, 2) \rangle$. How many elements are in G/H ? Is G/H Abelian? Is G/H cyclic?

$$|\langle (2, 2) \rangle| = \text{lcm}(4, 2) = 4. \quad |\mathbb{Z}_8 \oplus \mathbb{Z}_4| = 32 \quad |G/H| = 32/4 = 8.$$

• G/H is Abelian since G is Abelian

$$\langle (2, 2) \rangle = \{(2, 2), (4, 0), (6, 2), (0, 0)\}$$

• G/H is not cyclic. Consider $(a, b) + \langle (2, 2) \rangle$.

$(a, b)^2 = (2a, 2b) \in \langle (2, 2) \rangle$. Thus each non identity element has order 2. $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$.

37 **Exercise 2.** Let G be a finite group and let $H \triangleleft G$. Prove the order of $gH \in G/H$ must divide the order of g in G .

Let $|g| = k$. Then $(gH)^k = g^k H = eH = H$.

Since $(gH)^k = H$, then the order of gH must divide k .

Theorem. Let G be a group and let $Z(G)$ be the center of G . If $G/Z(G)$ is cyclic, then G is Abelian.

Proof:

Let $G/Z(G)$ be cyclic. Then $G/Z(G) = \langle gZ(G) \rangle$. Let $a \in G$. Then $(gZ(G))^k = aZ(G) \Rightarrow g^k Z(G) = aZ(G) \Rightarrow g^k z = a$. Note that b/c $g^k, z \in C(g)$, $a \in C(g)$

Thus each a in G commutes with g and $g \in Z(G)$.

Therefore $gZ(G) = Z(G)$ and $[G:Z(G)] = 1$,

implying $G = Z(G)$ and G is Abelian.

Question 8. Let G be a group of order 35 with a non-trivial center. Why must G be Abelian?

Possible orders of $Z(G)$ are 5, 7, and 35.

If $|Z(G)| = 5$, $[G:Z(G)] = 7 \Rightarrow Z(G)$ is cyclic $\Rightarrow G$ is Abelian.

If $|Z(G)| = 7$, $[G:Z(G)] = 5 \Rightarrow Z(G)$ is cyclic $\Rightarrow G$ is Abelian.

If $|Z(G)| = 35$, G is Abelian.

Theorem.

$$G/Z(G) \cong \text{Inn}(G).$$

Question 9. What can you say about $\text{Inn}(D_6)$?

$Z(D_6) = \{R_0, R_{180}\} \Rightarrow D_6/Z(G)$ has order $12/2 = 6$.

From Chap. 7, $D_6/Z(D_6) \cong D_3$ or Z_6 . If Z_6 , then by prev. thm, $D_6/Z(D_6)$ would be cyclic $\Rightarrow D_6$ is Abelian.

So it is isomorphic to D_3 .

Theorem (Cauchy's Theorem for Abelian Groups).

Let G be a finite Abelian group and let p be a prime that divides $|G|$. Then G has an element of order p .

21 **Exercise 3.** Let G be an Abelian group of order 33. Prove that G is cyclic.

Since 3 and 11 are primes that divide 33, G has elements p and q such that $|p|=3$ and $|q|=11$.

[show $|pq|=33$] Consider $pq \in G$. Then $|pq|=1, 3, 11$, or 33.

Let $k=|pq|$. Then $e=(pq)^k = p^k q^k$.

If $k=1$: $p^{-1} = q$ x

If $k=3$: $e = p^3 q^3 = q^3$ x

If $k=11$: $e = p^{11} q^{11} = p^{11} = p^2$ x

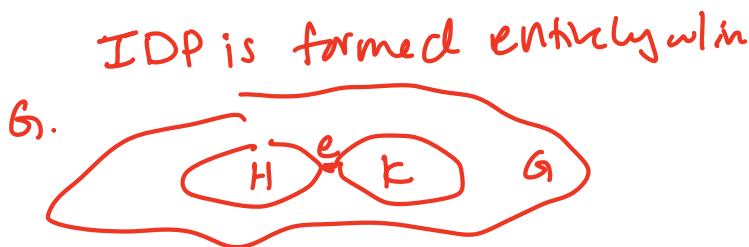
Thus $k=33$, and $G = \langle pq \rangle$.

Definition. A group G is the **internal direct product** of subgroups H and K ($G = H \times K$) if:

- H and K are normal subgroups of G .

- $G = HK$ $|HK| = \frac{|H||K|}{|H \cap K|}$

- $H \cap K = \{e\}$



The Internal Direct Product of two subgroups is nice because:

it is isomorphic to the external direct product.

EDP can be any 2 groups

Theorem.

If a group is the internal direct product of $H_1, H_2, \dots, H_n$,
 then $G \cong H_1 \oplus H_2 \oplus \dots \oplus H_n$.

Question 10. Why are each of the three conditions of IDP important?

Question 11. How many elements of order 6 are in $\mathbb{Z}_9 \times \mathbb{Z}_6$? How many cyclic subgroups of order 6 are there in this group?

BAD: migrated from Chapter 8.

Additional Exercises

1. Determine the order of $(\mathbb{Z} \oplus \mathbb{Z}) / \langle (4, 2) \rangle$. Is this group cyclic?

• $|\mathbb{Z} \oplus \mathbb{Z} / \langle (4, 2) \rangle|$ is infinite

• No: $(1, 1) + \langle (2, 2) \rangle$ has order 2. In an infinite cyclic group, all elements (nonidentity) have infinite order.

- 34 2. In $\mathbb{Z}$, let $H = \langle 5 \rangle$ and $K = \langle 7 \rangle$. Does $\mathbb{Z} = HK$? Does $\mathbb{Z} = H \times K$?

- $\gcd(5, 7) = 1$, and $1 = 3(5) - 2(7)$. Since $1 \in HK$, $\mathbb{Z} = HK$.
- No. $\mathbb{Z} \neq H \times K$ b/c $H \cap K \neq \{e\}$ since $35 \in H \cap K$.

Recall: $HK = \{hk : h \in H \text{ and } k \in K\}$

3. Let p be a prime and G be a group of order $2p$. Show that if H is a subgroup of G that is not normal, then H has order 2. (contrapositive)

Let H be a subgroup of G , where $|G| = 2p$. Suppose that $|H| \neq 2$. Then either ① $H = \{e\}$, ② $H = G$, or ③ $|H| = p$.

If $H = \{e\}$ or $H = G$, H is clearly normal in G .

If $|H| = p$, then $[G:H] = 2$. Thus $H \triangleleft G$.

4. Let H be a normal subgroup of G and let $a \in G$. If the element aH has order 3 in the group G/H and $|H| = 10$, what are the possibilities for the order of a ?

- We know $|aH|$ divides $|a|$ in G .
- B/c $|aH| = 3$, $a^3 \in H$. Thus $|a^3|$ divides 10
and $|a^3| = 1, 2, 5, 10$. $\Rightarrow a^3 = e, a^6 = e, a^{10} = e$ or $a^{30} = e$.
- So possible orders of a are 3, 6, 15 or 30.

5. Prove that an Abelian group of order 63 is cyclic. $63 = 9 \cdot 7 = 3^2 \cdot 7$.

WRONG! $\mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_7$ is not cyclic.

6. Prove that D_4 cannot be expressed as the internal direct product of two proper subgroups.

Suppose $D_4 = H \times K$. Because $|D_4| = 8$, then wlog, $|H| = 4$ and $|K| = 2$. Thus H & K are both Abelian groups.

Also $D_4 \cong H \oplus K$. Since H & K are both Abelian, D_4 must also be Abelian, a contradiction.